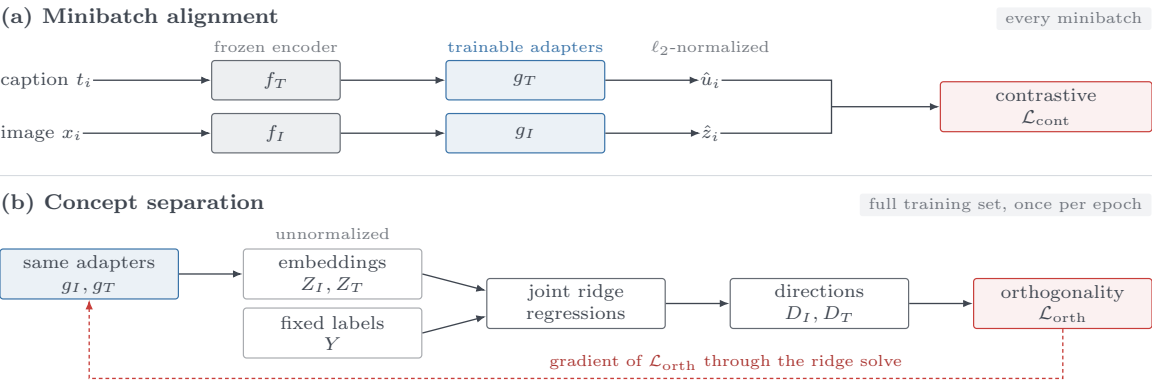

(a) Minibatch alignment
every minibatch
frozen encoder
trainable adapters
ℓ_2-normalized
caption t_i
image x_i
f_T
f_I
g_T
g_I
$\hat{u}_i$
$\hat{z}_i$
contrastive
$\mathcal{L}_{\text{cont}}$
(b) Concept separation
full training set, once per epoch
unnormalized
same adapters
g_I, g_T
embeddings
Z_I, Z_T
fixed labels
Y
joint ridge
regressions
directions
D_I, D_T
orthogonality
$\mathcal{L}_{\text{orth}}$
gradient of $\mathcal{L}_{\text{orth}}$ through the ridge solve